

STRATHMORE INSTITUTE OF MATHEMATICAL SCIENCES

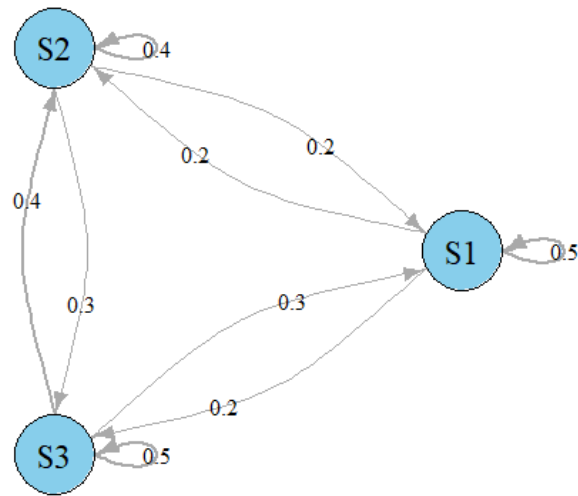
Master of Science (Data Science and Analytics)

STA 8405- PROBABILITY AND STOCHASTIC PROCESSES

Practice questions

- 1) Clearly, distinguish the following terms
 - i. Stochastic Process and Filtration
 - ii. Super Martingale and Sub-Martingale
 - iii. Single channel-single phase system and Single channel-multiphase system
 - iv. Multi-channel – Single phase system and Multi-channel – multi phase system
 - v. σ -algebra and Hidden Markov Chain
 - vi. Higher-order Markov chains and Time-homogeneity Markov chain
 - vii. Counting Process and Branching Process
 - viii. Communication class and Absorbing class
 - ix. Transient state and Recurrent state
 - x. Irreducible and Reducible Markov chain
 - xi. Ergodic and Non-ergodic Chain
 - xii. Limiting distribution of Markov chain
 - xiii. Gambler's Ruin and Renewal Process
 - xiv. Periodicity and Aperiodicity
 - xv. Brownian Motion and Random Walk
- 2) Show that for a typical random walk, the mean is given by $E(S_n)=n(2p-1)$ and variance $Var(S_n) = 4np(1 - p)$
- 3) Describe the Markovian property
- 4) Derive the formulas for the Gambler's Ruin Problem for situation when $p=q$ and $p \neq q$. Also describe how to become infinitely rich or getting ruined in the same Gambler's Ruin setting
- 5) Provide 5 components of A hidden Markov model
- 6) Tom starts with \$5, and $p = 0.63$: What is the probability that Tom obtains a fortune of $N = 12$ without going broke? What is the probability that Tom will become infinitely rich? If Tom instead started with $i = \$2$, what is the probability that he would go broke?
- 7) Collins bought a share of stock for \$10, and it is believed that the stock price moves (day by day) as a simple random walk with $p = 0.6$. What is the probability that Collins' stock reaches the high value of \$25 before the low value of \$4?
- 8) Patrick bought a share of stock for \$10, and it is believed that the stock price moves (day by day) as a simple random walk with $p = 0.55$. What is the probability that Patrick's stock reaches the high value of \$15 before the low value of \$5?
- 9) Provide three (3) assumptions of Hidden Markov Models
- 10) What are the three assumptions of Pure Birth Process (Yule-Furry Process)
- 11) With relevant Kolmogorov Forward Equations (Master Equations), show that the equation for the mean of a Pure Birth and Death Processes is given as
$$m(t) = m(0)e^{(\lambda-\mu)t}$$
- 12) State and use the four Chapman-Kolmogorov Postulates to derive the Poisson Process. Hence show that the mean and variance is given as λt .
- 13) Study the Markov Chain graph below and answer the following questions

3-State Markov Chain



- Generate the transition matrix
 - Define and state the communication class
 - Clearly provide concrete reasons why above chain can be considered as an Ergodic Chain
 - Do we expect limiting distribution to be achieved in this chain? Why?
 - State the formula for a transition matrix in a)
- 14) Consider a Markov chain with two possible states, $S = \{0, 1\}$. Suppose that the transition matrix is given by

$$P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}$$

where a and b are two real numbers in the interval $[0, 1]$ such that $0 < a + b < 2$. Using induction (or any other method), show that

$$\lim_{n \rightarrow \infty} P^n = \frac{1}{a+b} \begin{bmatrix} b & a \\ b & a \end{bmatrix}$$

- 15) Consider M/M/1: This is the simplest queuing system, where the arrival is considered Poisson Process and service mechanism, Exponential Process. The system consists of only one server, one unlimited FIFO queue and unlimited customer population. Derive
- Average entities in the system (L_s)
 - Average entities in the queue (L_q)
 - Average Waiting time in the system (W_s)
 - Average Waiting time in the queue (W_q)
- 16) Explain the three components of Components of a Queuing System/model
- 17) A telephone attendant receives 110 calls during the busy hour. Each call takes, on average, 30 seconds to process.
- What percentage of the attendant's time is devoted to answering calls?
 - How long must people wait, on average, before their call is processed?
 - How many people are in the queue, on average?